IE199 – F04

جامعة المستقبل
Mustaqbal University

College of Engineering & Computer Science

Industrial Engineering Department

Designing and Manufacturing a Protective and Interactive Helmet for Children.

Introduction to Engineering and Design [IE-199]

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ABSTRACT

The main objective is to design and manufacture an protective, interactive, and therapeutic helmet for children. The helmet aims to ensure their safety during activities and address deformities. The project required a variety of materials, including software (SolidWorks and Ultimaker Cura), manufacturing methods (3D printer (Ender 3 V2), and materials (PLA), and an Electric circuit (Arduino UNO).As a result, we finally achieved in designing an protective, interactive and therapeutic helmet to protect a child's head. The helmet features a specific LED light that illuminates on the affected side upon detecting a fall for 10 minutes, providing a clear visual indicator to notify parents of their children's falls and deliver immediate feedback on the impact.. Also, it promotes proper cranial development.

INTRODUCTION

In this fast-developing era, it is always unpredictable when, where, to whom and how accidents might occur but the consequence of it might cause injuries of various severity or even death. As an additional problem that we need to solve, the baby's head during the first year is very sensitive. Because of that, our project is to design and manufacture a protective, interactive and therapeutic helmet for children under 12 years old. The meaning of interactive is to let the parents know if their children fall, and to provide real-time feedback on the impact and severity of the fall. Additionally, the helmet will be designed to promote proper head positioning during sleep and provide gentle support to prevent any misshaping or deformities.

OBJECTIVES

1. Design and manufacture a protective, interactive, and therapy helmet for children to ensure their safety and health while engaging in activities by the end of the 13th week.
2. The helmet's design should be recyclable, long-lasting, and provide optimal protection and comfort.
3. We seek to reduce the cost so that more people can afford our product.
4. Design a helmet that combines protection and treatment so that it corrects the shape of the child's head.

MATERIALS & METHODS

- **SolidWorks:** a 3D computer modeling software that has been used to design all the models and simulate the assembly process accurately and provide a visual presentation of the project.
- **Cura:** a 3D slicer software that prepares the models designed in SolidWorks to be 3D printed using a 3D printer
- **Ender 3 Pro:** a 3D printer that is used to manufacture the designed parts of the project out of plastic with great accuracy.

Fig 1: Pictures of all the methods and software used to complete this project

RESULTS & DISCUSSION

Statistics showed the importance of manufacturing a helmet that protects the child's head, so a helmet was manufactured and designed to protect the child's head from the risks of falling, and also interactive through LEDs that work for 10 minutes when one of the pushbuttons is pressed by falling on the head, the helmet can also modify the shape of his head.

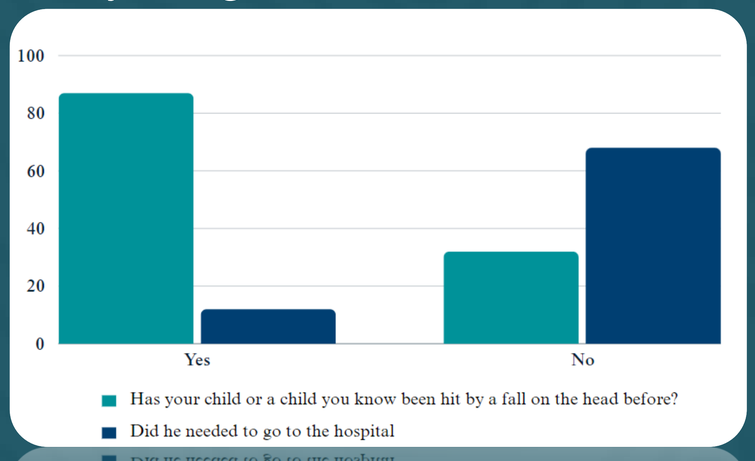

Fig 2: A graph showing the percentage of 200 submissions for children admitted to the hospital after a fall.

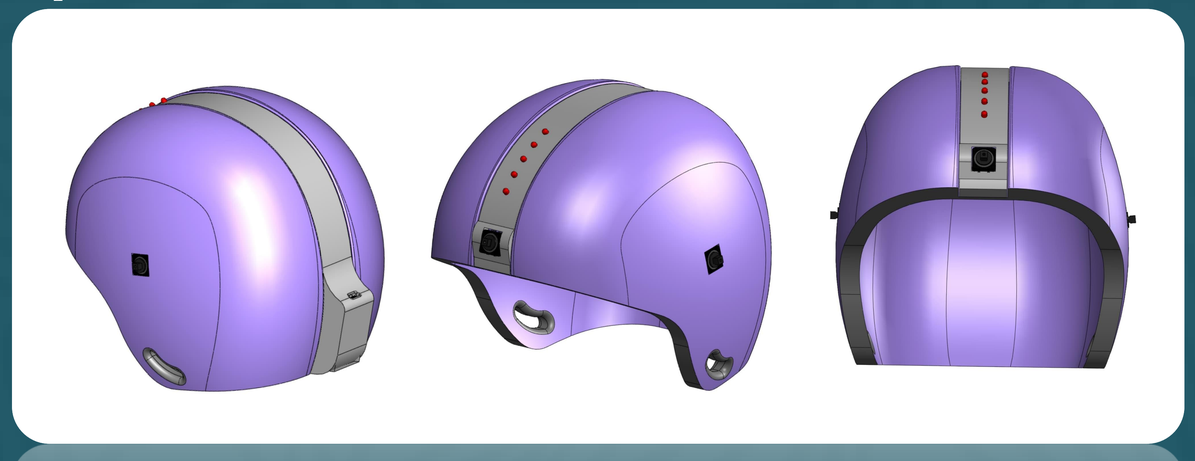

Fig 3: Pictures of the final protective helmet design

CONCLUSION

The aim of this project was to produce a protective, interactive, and therapeutic helmet specifically tailored for children and provide a secure environment that fosters their well-being.

REFERENCES

[1] W. Al Mehaileb,S. Al Mutairi .2030. "Designing and Manufacturing a Protective and Interactive Helmet for Children". Mustaqbal University, Saudi Arabia.